Ensembling method (Group of models put together)



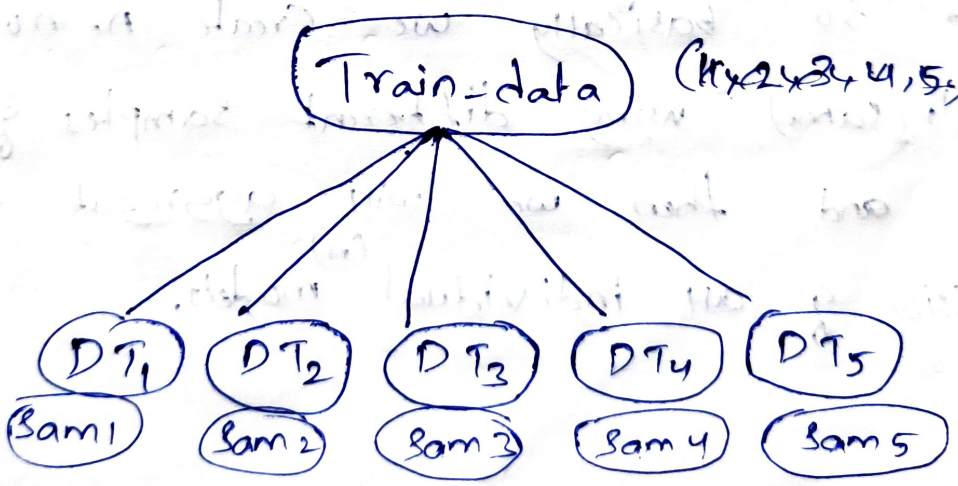
Bagging (Same - model, different data)

Boosting (one-model, one-data)

Stacking. (different model, same data)

In Bagging we use a concept called Bootstrap Sampling.

Here Boolstrap sampling is nothing but sampling with replacement technique.



We need to understand the sampling process this happens in 2-ways

(1) Row_sampling.

(2) Column - Sampling.

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→ Row Sampling :- We need to decide the max_sample. / No. of Rows.

→ Column Sampling :-

Here we need to decide the max_features / No. of Columns basically it is Sqrt (Total features)

log (Total features).

9 - col ⟶ √9 - 3 col will be chos

Now let's understand how the model works so basically we create n. no of models (same) using different samples of data and then we will aggregate the decision of all individual (n) models.

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→ Assume the training is finished

→ Now a new test data point will come and assume the no of estimators are (5 knn) so total we have 5 KNN models so now the test data will go through all the KNN's and in total we will have 5 KNN's predichings the test data and we will take majority of the predichions (mode).

→ This sort of prediction will be more accurate than a single model.

→ It's like prediction from multiple models is better than single model.

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Advantages of Bagging.

→ Instead of (1) we will take cummulative decision.

→ The Variance error will be reduced.

→ By using Bagging technique multiple features will be used.

Bagging:- Important points.

(1) Group of same model.

(2) Data selection
- Row sampling :- with replacement
- Column sampling :- -1, log, sqrt.

(3) Bagging is a parallel process.

(4) Bagging reduces Variance Error.

(5)

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⑧ Random Forest :-

(Bagging with Decision tree).

→ Random Forest is Group of D.T's put together.

→ Basically Decision trees are very greedy because the root node split will always be the topmost same feature but by using Random forest, because by column sampling the DT will have different Root Nodes.

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⑧. Decision Tree / Random Forest

Regressor / Regressor:

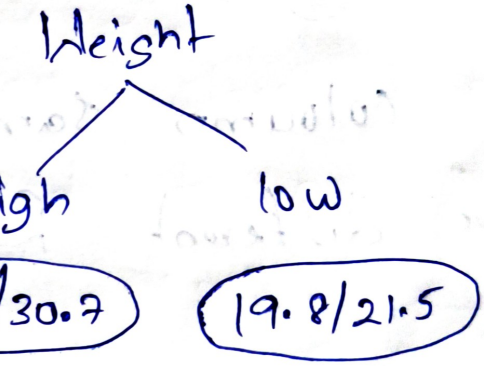
→ How do we calculate Root Nodes

↓↓

Gini, Entropy ⇒ classification.

Root Node → MSE ⇒ Regressor.

Weight	bmi
high	28.7
high	26.2
low	19.8
low	21.5
high	30.7



MSE:- 2.5 MSE = 1.2

$$MSE = \sum (x - \bar{x})^2 / n$$

$$\text{Total MSE} = \frac{3}{5} \times (2.5) + \frac{2}{5} \times (1.2)$$

Total gini = $P(H)\,G(H) + P(L)\,G(L)$ (CAT)

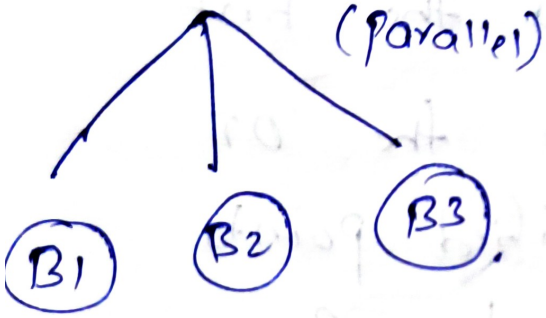
⇒ $P(H)\,MSE(H) + P(L)\,MSE(L)$ (CON).

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⑧. Boosting.

→ Basically Boosting is a SEQUENTIAL process. where as Bagging is parallel process.

Bagging.



Boosting.

(Sequential)



Total 3 boosting techniques.

① Ada Boosting

② Gradient Boosting

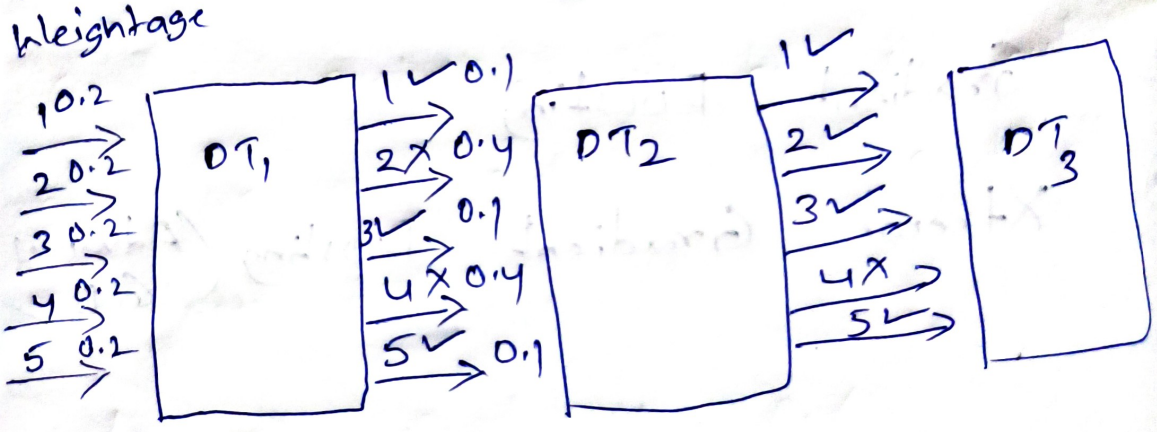
③ Xtreme Gradient Boosting (Advanced GB)

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⊛. Ada Boost. (Group of Decision tree)

→ ~~Initially in Ada Boost all the algorithm~~

→ Initially in Ada Boost all the data points will be passed to the first DT model and then from the DT predictions all the misclassified points will be given high weightage (more importance) than correctly classified points and then again passed to DT_2 this process continues.



⊛ After how many iterations Ada boost stops. (It depends on the error, as long as the error is negligible / no. of estimators)

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Initial Weights will be equal.

After 1st Iterations Weight will be updated.

New Weights for Wrongly classified data points
= Sample Weight $\times e^{\text{amount of say}}$

New Weights for rightly classified data points.
= Sample Weight $\times e^{-\text{amount of say}}$.

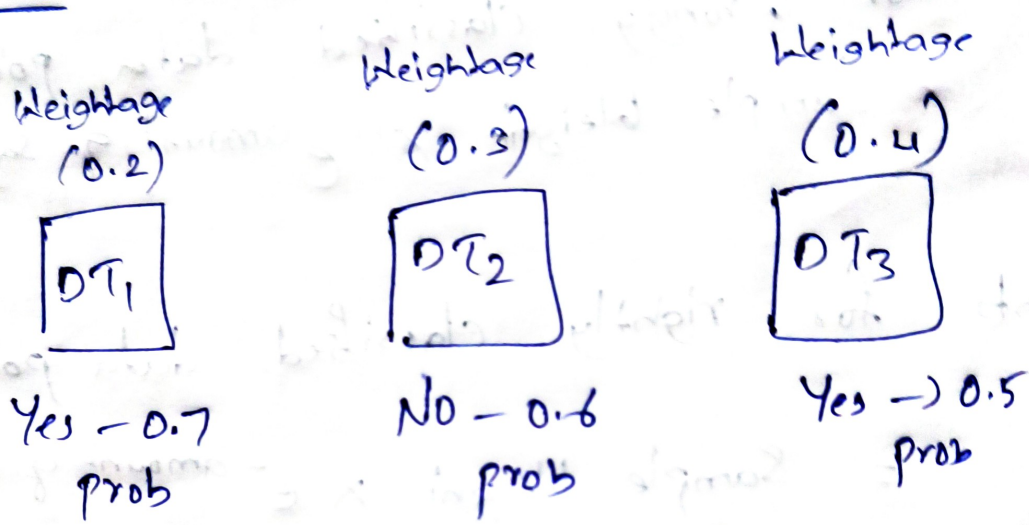
Amount of say $\frac{1}{2} \ln\left(\frac{1 - \text{Total error}}{\text{Total error}}\right)$

Total error :- $\frac{\text{Wrong classified data}}{\text{All data points}}$

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→ In Ada Boost the training happens sequentially but the testing happens parallely.

Testing:-



o/p here the majority is yes 2 model- yes
1 model- no
so the prediction will be yes but here we also need to look in to the weightages.

Yes → 0.2 + 0.4 = 0.6

No → 0.3 = 0.3

Now, the catch the the deeper Decision tree will be given the highest weightage and that will be the correct classification most of time.

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